

G.C.E. Advanced Level Physics

Mechanics - Core Exam Notes

Special Exam Point: Conceptual Clarity

The First Law qualitatively defines *inertia* and introduces the concept of **inertial reference frames**—frames of reference in which Newton's laws hold true without fictitious forces. An object moving with constant velocity or at rest has a resultant force $\Sigma F = 0$ (Translational Equilibrium).

Crucial A/L Examination Pitfalls on Action-Reaction Pairs

Students frequently confuse action-reaction pairs with balancing forces. To avoid losing marks in structured essay questions, remember these four golden rules of action-reaction pairs:

- They act on **two different bodies** (never on the same body).
- They are of the **exact same nature** (e.g., both gravitational, both electrostatic, or both contact normal forces).
- They occur simultaneously.
- **Example:** A book resting on a table. The gravitational pull on the book by the Earth and the normal reaction from the table are *not* action-reaction pairs (they act on the same body and result in equilibrium). The true action-reaction pair to the Earth pulling the book is the book pulling the Earth upward with an equal gravitational force.

5. Summary of Key Applications in A/L Problems

Physical Scenario	Governing Equation / Approach	Key Exam Consideration
Lifts / Elevators	$R - mg = ma$ (accelerating upwards)	Apparent weight (R) changes with acceleration; zero in free fall.
Connected Bodies (Pulleys)	$T - mg = ma$ or $mg - T = ma$	Assume strings are inextensible and light; pulleys are frictionless and smooth unless specified.
Inclined Planes	$mg \sin(\theta) - f = ma$ (along plane) $R = mg \cos(\theta)$ (perpendicular)	Resolve weight components parallel and perpendicular to the incline.